

Fall Detection: Progress Presentation

Abbos Aliboev

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Motivation



Construction sites are dangerous

- Falls are the No-1 cause of death on construction sites
- Many falls go unseen – no one reacts in time



Real-time detection saves lives

- Stop machines, call help immediately



Why vision-based?

- Cameras are cheaper, always on
- Sensors are limited, uncomfortable
- Challenge: occlusion & speed

Problems in Existing Work

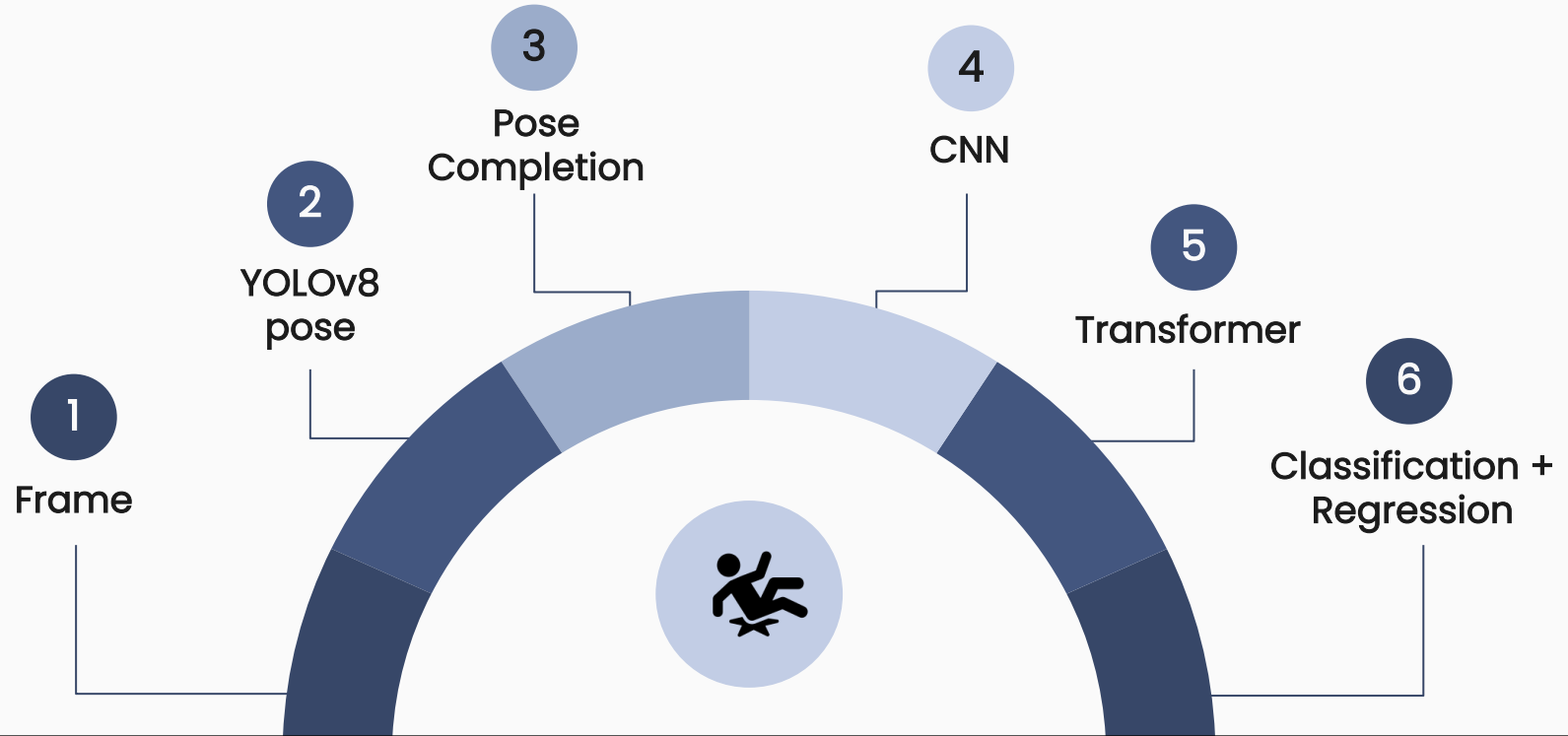
- 1 Sensor-based systems: expensive, non-scalable
- 2 Old vision models: slow, not real-time
- 3 Occlusion not handled
- 4 Pre-fall not predicted

Our Objective

- 1 Real-time detection
- 2 Handle occlusion
- 3 Predict fall before it happens (Pre-fall)
- 4 Estimate time-to-fall (regression)
- 5 Edge-compatible (Jetson, etc.)



System Overview Diagram



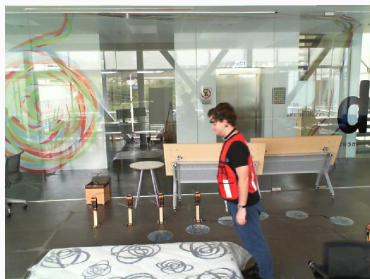
Dataset Use

UP-Fall – only frames

Publicly available

17-folders, one per subject:

11-subfolders, one per activity:



Custom video dataset

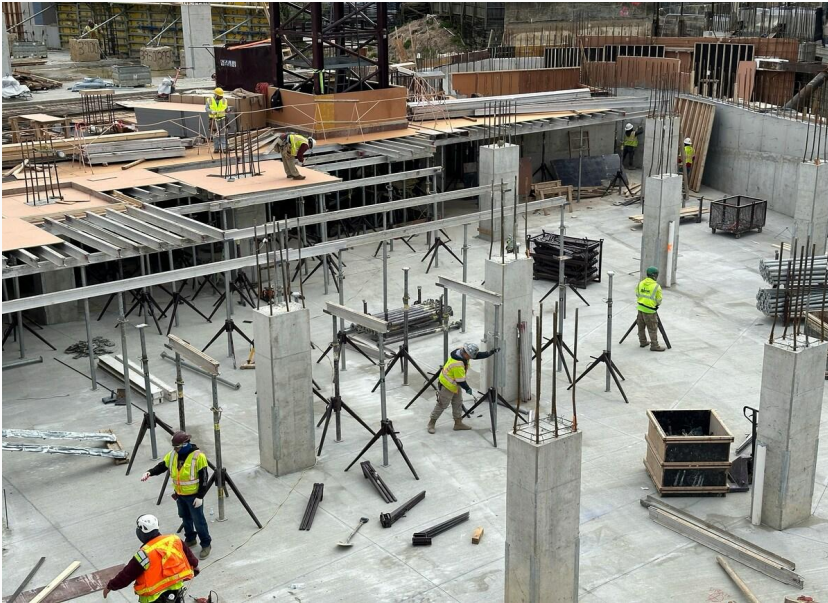
Multiple people in a frame (multi-person scenarios)

PPE gear (helmets, vests, etc.)

Various fall types with occlusion

Pre-fall behavior recorded explicitly

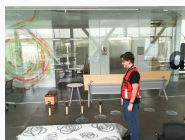
Pose Completion



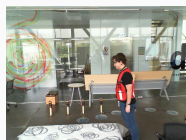
Problem: Occlusion = missing joints $\Rightarrow$ **Solution:** MLP or AutoEncoder fills missing keypoints $\Rightarrow$ Lightweight for real-time

Sequence Modeling (CNN)

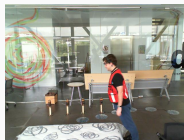
- ✓ Detects short-term motion patterns
- ✓ Works on a sliding window of pose vectors
- ✓ Very fast – ideal for real-time systems



Frame 1



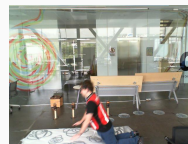
Frame 2



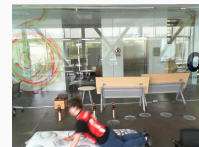
Frame 3



Frame 4



Frame 5



Frame 6



Frame 7

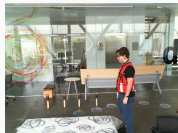
Used as a motion window for CNN

Temporal Attention (Transformer Encoder)

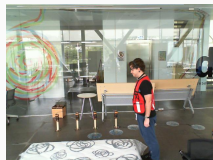
Captures important frames in time

Focuses on motion before fall

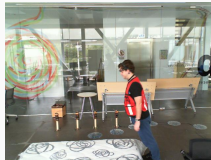
Efficient attention head



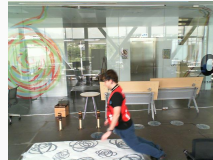
Frame 1



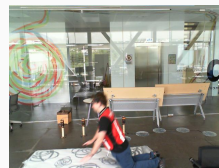
Frame 2



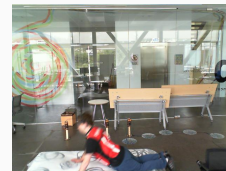
Frame 3



Frame 4



Frame 5



Frame 6



Frame 7

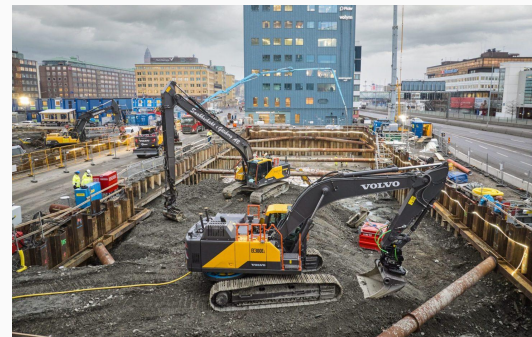
Dual Output Head

Output 1	Output 2
Classification	Regression
Fall / Pre-fall / No fall	Time-to-fall in seconds (tff = 1.45)

Fall tff = 1.45



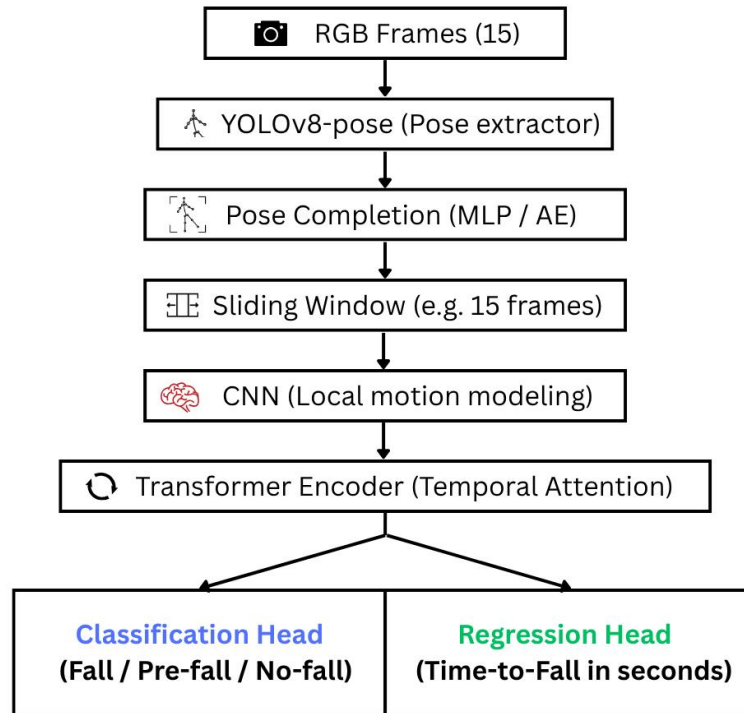
Dual Why Is Regression Important?



Regression predicts *"how many seconds until a fall occurs"* - *(e.g., "1.7 seconds to fall")

- Enables early warning systems: **alarm, drone, emergency response**
- Helps take action before the actual fall happens
- Offers a precise estimate of how close the person is to falling

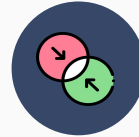
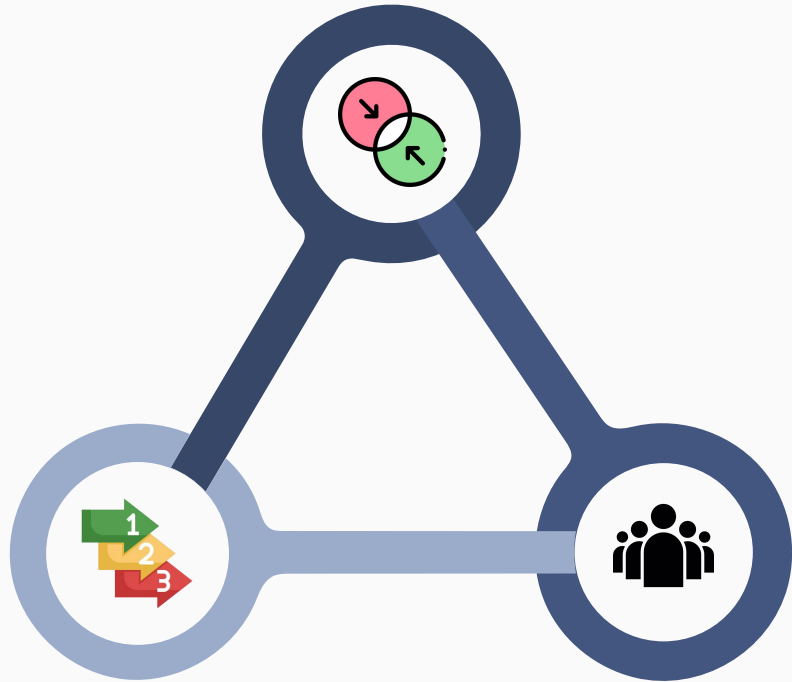
Model Architecture Overview



Comparative Table

Model	Occlusion	Pre-fall	Regression	Attention	Real-time
YOSAP	✗	✓	✗	✗	✗
TCNTE	✗	✗	✗	✓	✓
TST	✗	✗	✗	✓	✗
Ours	✓	✓	✓	✓	✓

Model Innovation Summary



First

Combine pose completion +
attention + TTF regression



Multiple people

Multi-person ready, real-time capable



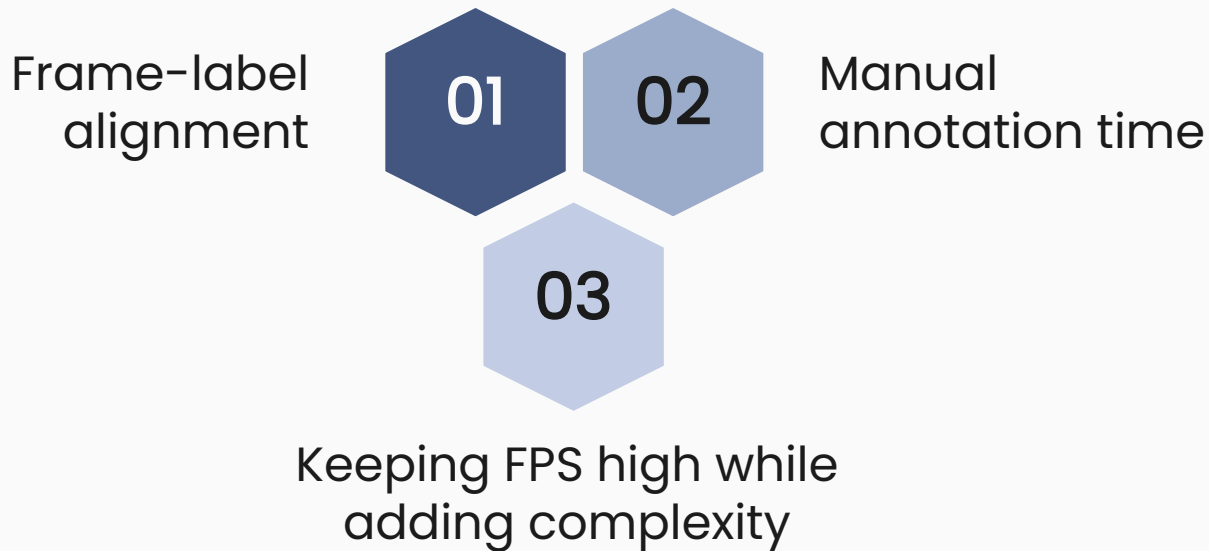
3 Stage fall detection

Fall prevention, not just detection

Development Progress

Task	Status
Dataset exploration	In progress (analyzing UP-Fall, reviewing options)
Pose extraction (YOLOv8-pose)	Planned
Pose completion	Studying techniques (MLP, AutoEncoder)
Annotation (Pre-fall, TTF)	To be designed
Model structure comparison	CNN, TCN, Transformer analysis ongoing
Training	Not started
Evaluation	Not started

Technical Challenges



Conclusion & Final Thoughts

We proposing a **real-time vision-based fall detection system** tailored for **construction site environments**

Our model handles:

- **Occlusion** via Pose Completion
- **Short-term motion** via CNN
- **Temporal patterns** via Transformer
- **Fall risk** via Classification + Regression

 *Currently under development – focusing on:*
→ *Annotation, model training, real-time optimization*

Thanks!

Do you have any questions?

www.abbosaliboev.site

Arxitektura	FPS (real-time?)	Spatial info	Temporal context	Occlusion handling	Multiple people	Generalizability
CNN + Transformer	⚠ Medium	✓	✓	✗	✗	⚠ (data heavy)
TCN + Transformer	✗ Sekin	✗	⚠ (cheklangan)	✗	✗	⚠
ST-GCN	✗ Sekin	✓ (graph)	✓	⚠	✓	✗ (pose only)
YOLO + Pose + Transformer + TTF (biz)	✓ Optimized	✓ (bbox+pose)	✓ (deep attention)	✓ (pose complete)	✓	✓